Mohammed VI Polytechnic University (UM6P)

Bachelor

Numerical Analysis Lab

Practical Work

Numerical Solution of Ordinary Differential Equations

Exercice 1:

Problem Statement

We consider the differential equation:

$$\begin{cases} y'(t) = f(t, y(t)) = -t\, y^2(t), \quad t > 0, \\ y(0) = 2 \end{cases}$$

Exact Solution

The exact solution is:

$$y(t) = \frac{2}{1+t^2}.$$

Numerical Methods

Forward Euler Method (Explicit)

The Forward Euler scheme in derivative form:

$$\frac{u_{n+1} - u_n}{h} = f(t_n, u_n), \quad n = 0, 1, \dots, N-1, \quad u_0 = 2$$

Which gives:

$$u_{n+1} = u_n + hf(t_n, u_n) = u_n - ht_n u_n^2$$

Backward Euler Method (Implicit)

The Backward Euler scheme in derivative form:

$$\frac{u_{n+1} - u_n}{h} = f(t_{n+1}, u_{n+1}), \quad n = 0, 1, \dots, N-1, \quad u_0 = 2$$

Substituting $f(t_{n+1}, u_{n+1}) = -t_{n+1} u_{n+1}^2$:

$$\frac{u_{n+1} - u_n}{h} = -t_{n+1} u_{n+1}^2 \quad \Rightarrow \quad u_{n+1} = u_n - ht_{n+1} u_{n+1}^2$$

Tasks for Students

1. Compute u_n for $n = 0, 1, \ldots, 20$ using Forward Euler.
2. Compute u_n for $n = 0, 1, \ldots, 20$ using Backward Euler.
3. Compute the exact solution $y(t_n)$ at the same points.
4. Plot the exact solution and both numerical solutions on the same figure.
5. Comment on the obtained results in the figure.

Exercice 2:

Problem Statement

We consider the Cauchy problem:

$$\begin{cases} y'(t) = f(t, y(t)) = -y(0.1 - \cos(t)), \quad t > 0, \\ y(0) = 1 \end{cases}$$

Exact Solution

The exact solution is:

$$y(t) = e^{-0.1t + \sin(t)}$$

Numerical Methods

Forward Euler Method (Explicit)

The Forward Euler scheme in derivative form:

$$\frac{u_{n+1} - u_n}{h} = f(t_n, u_n), \quad n = 0, 1, \ldots, N-1, \quad u_0 = 1$$

Which gives:

$$u_{n+1} = u_n + hf(t_n, u_n) = u_n - h\,u_n\,(0.1 - \cos(t_n))$$

Heun Method (Improved Euler)

The Heun method:

$$u_{n+1} - u_n = \frac{h}{2}\big(f(t_n, u_n) + f(t_{n+1}, u_n + hf(t_n, u_n))\big), \quad n = 0, 1, \ldots, N-1$$

Parameters

$$\text{Interval: } [0, 12], \quad h = 0.4, \quad N = \frac{12}{0.4} = 30$$

$$t_n = nh, \quad n = 0, 1, \ldots, 30$$

Tasks for Students

1. Compute u_n for $n = 0, 1, \dots, 30$ using Forward Euler.
2. Compute u_n for $n = 0, 1, \dots, 30$ using Heun method.
3. Compute the exact solution $y(t_n)$ at the same points.
4. Plot the exact solution and both numerical solutions on the same figure.
5. **Which method is more accurate? Justify your answer based on the figure.**
6. **Compute the Forward Euler solution for different time steps:** $h = 0.4, 0.2, 0.1, 0.05$. Compare how the solution changes when decreasing h.
7. **Plot the solutions for these different** h **values and compare each with the exact solution.** Discuss how the solution improves as h decreases.